PA2 Overview

PA2

- **Goal**
  - Final goal is to create a shell with signal, redirection and pipe support.
  - This assignment can be divided into four parts:
    - » Implementing 6 executables found in GNU Core Utilities (coreutil), which are head, tail, cat, cp, mv, and rm.
    - » Extending shell by using concepts from weeks 5~7, including signal handling, process groups, redirection.
    - » Adding pipe support
    - » Writing a report on your work

PA2

- **Restrictions**
  - This is a personal project. You can discuss the task together, but you must write the source code on your own.
  - Use the Linux system call/library functions you have learned so far to implement the task.
    - » This means that you can use printf(), readline(), etc. as well (not just read(), write()).
    - » You can also use other standard POSIX and GNU C functions/extensions like getopt(), argp(), or error() if needed.
    - » However, use of system() or popen()/pclose() functions is prohibited.
  - If a resource is dynamically allocated, it must be freed before the program terminates.
    - » Resources refer to files, memory, and child processes.
  - If the pgid of the child process created by the shell is the same as the pgid of the shell, the submission is not scored.

PA2 Organization

- **pa2 should be organized in the following manner**
  - pa2/
    - » Makefile
    - » executable_src/*.{c,h} (contains pa2_cp.c, etc.)
    - » shell_src/*.{c,h} (contains files related to pa2)
    - » **build/** (should be removed when submitting pa2)
      - » Contains object files
    - » **bin/** (should be removed when submitting pa2)
      - » **pa2**, pa2_head, pa2_tail, pa2_cat, pa2_cp, pa2_mv, and pa2_rm
  - When *make* is called, it should compile all the binaries (pa2_cp, pa2_mv, **pa2**, etc.) to bin/
  - However, when bundling pa2, **bin/** and **build/** should be removed.
    - » i.e. tar file should only include Makefile, executable_src, and shell_src

PA2 Submission

- **Submit two files in iCampus**
  - Bundled tar file with *source code* and Makefile

    ```
    $ tar -caf STUDENTID_pa2.tar.zst pa2
    ```

    » We'll grade your submission with **make**

    » If compilation fails, your points for this exercise will be zero
  - Report (**.pdf**)

    » Should contain the project design and implementation details.
- **Ten points will be deducted per day if submitted past the deadline. (max 4 days)**
- **Worth 100 points**
- **Due date: May 3, 2026 (23:59)**